

AFS (Air Facility System) Data Dictionary

Source: EPA, *AFS Data Download* documentation (April 2014); ECHO AFS archive, <https://echo.epa.gov/tools/data-downloads#afs>.

Scope. AFS is the pre-2014 Air Facility System; the ECHO download is **frozen as of October 17, 2014**. Data are rolled up to the plant level. Each plant joins on **AFS_ID** (also called SCSC), a 10-character code (Census FIPS state + county + AFS plant number). AFS does not carry a **REGISTRY_ID**, so there is no built-in crosswalk to ICIS-Air's **PGM_SYS_ID**. Field names, types, and code labels below follow EPA's AFS documentation; field names and the *observed* code values have been verified against the actual downloaded CSVs (discrepancies noted).

AFS_FACILITIES.csv

One row per plant: identification, location, classification, operating, and compliance status.

Field	Description
PLANT_ID	Numeric plant identifier
AFS_ID	10-character plant code (state FIPS + county FIPS + AFS plant number); also "SCSC"
PLANT_NAME	Plant name
EPA_REGION	EPA region (01–10)
PLANT_STREET_ADDRESS / PLANT_CITY / PLANT_COUNTY / STATE / STATE_NUMBER / ZIP_CODE	Location fields
PRIMARY_SIC_CODE / SECONDARY_SIC_CODE	Four-character Standard Industrial Classification codes
NAICS_CODE	Six-character NAICS code
AFS_GOV_FACILITY_CODE	Government facility indicator (see codes below)
FEDERALLY_REPORTABLE	Y/N; ECHO-generated (see definition below)
EPA_CLASSIFICATION_CODE	Emissions classification (see codes below)
OPERATING_STATUS	Operational condition (see codes below)
EPA_COMPLIANCE_STATUS	EPA compliance determination (see codes below)
CURRENT_HPV	Current High Priority Violator status (see codes below)
LOCAL_CONTROL_REGION	Two-character local control region code (meanings vary by state)
STATE_COMPLIANCE_STATUS	State agency compliance determination (same code set as EPA_COMPLIANCE_STATUS)

EPA_CLASSIFICATION_CODE (Alabama Power decision / 1993 EPA guidance)

Code	Description
A	Actual or potential emissions above the applicable major-source thresholds
A1	Actual or potential controlled emissions >100 tons/year
A2	Actual emissions <100 tons/year, but potential uncontrolled emissions >100 tons/year
B	Potential uncontrolled emissions <100 tons/year
C	Unregulated pollutant, actual or potential controlled emissions >100 tons/year
ND	Major-source thresholds not defined
SM	Synthetic minor: potential emissions below all applicable major-source limits
UK	Unknown classification

Correction: C is not “Unknown” (only UK is). Observed in the data: A, A1, A2, B, C, ND, SM, UK.

OPERATING_STATUS

Code	Description
O	Operating
C	Under Construction
P	Planned (applied for a construction permit)
T	Temporarily Closed
X	Permanently Closed
I	Seasonal
D	NESHAP Demolition
R	NESHAP Renovation
S	NESHAP Spraying
L	Landfill

Observed in the data: O, C, P, T, X, I (the NESHAP/landfill statuses D, R, S, L are documented but do not appear in this download).

EPA_COMPLIANCE_STATUS (worst case across EPA and state fields)

Code	Description
0	Unknown compliance status
1	In Violation – No Schedule
2	In Compliance – Source Test
3	In Compliance – Inspection
4	In Compliance – Certification
5	Meeting Compliance Schedule
6	In Violation – Not Meeting Schedule
7	In Violation – Unknown With Regard To Schedule
8	No Applicable State Regulation
9	In Compliance – Shut Down
A	Unknown With Regard To Procedural Compliance
B	In Violation With Regard To Both Emissions And Procedural Compliance

Code	Description
C	In Compliance With Procedural Requirements
D	HPV Violation (auto-generated)
E	FRV Violation (auto-generated)
F	HPV On Schedule (auto-generated)
G	FRV On Schedule (auto-generated)
H	In Compliance (auto-generated)
M	In Compliance – CEMs
P	Present, See Other Program(s)
U	Unknown By Evaluation Calculation
W	In Violation With Regard To Procedural Compliance
Y	Unknown With Regard To Both Emissions And Procedural Compliance

Codes A, B, C, P, U, W, Y appear in the data and are defined in EPA’s AFS documentation; a prior draft listed only 0–9, D, E, F, G, H, M.

CURRENT_HP

Code	Description
S	Violation Unaddressed; State/Local has lead enforcement
T	Violation Addressed; State has lead enforcement
E	Violation Unaddressed; EPA has lead enforcement
F	Violation Addressed; EPA has lead enforcement
B	Violation Unaddressed; EPA and State share lead enforcement
C	Violation Addressed; EPA and State share lead enforcement
X	Violation Unaddressed; enforcement lead unassigned

Observed in the data: C, E, F, S, T (B and X are documented but do not appear).

AFS_GOV_FACILITY_CODE

0 Privately owned/operated | 1 Federal | 2 State | 3 County | 4 Municipality | 5 District | 6 Tribe.
(All 0–6 observed.)

FEDERALLY_REPORTABLE

ECHO-generated Y/N. Per EPA documentation, “Y” if `EPA_CLASSIFICATION_CODE` $\in$ {A, A1, A2, SM}, OR (`AIR_PROGRAM_CODE` $\in$ {8, 9} and `EPA_COMPLIANCE_STATUS` $\neq$ 8).

AIR_PROGRAM.csv

One row per plant–program (with pollutant-level detail). Joins to facilities on `AFS_ID/PLANT_ID`.

Field	Description
AFS_ID / PLANT_ID	Plant identifiers
AIR_PROGRAM_CODE	One-character regulatory air program (see codes below)
AIR_PROGRAM_STATUS	Operating status within the program (same codes as OPERATING_STATUS)
EPA_CLASSIFICATION_CODE	Emissions classification at the air-program level
EPA_COMPLIANCE_STATUS	Compliance status at the air-program level
AIR_PROGRAM_CODE_SUBPARTS	Applicable subparts, space-delimited
POLLUTANT_CODE	Pollutant code at the air-program level
CHEMICAL_ABSTRACT_SERVICE_NMBR	CAS number for the pollutant, if any
POLLUTANT_CLASSIFICATION	Emissions classification at the pollutant level
POLLUTANT_COMPLIANCE_STATUS	Compliance status at the pollutant level

AIR_PROGRAM_CODE (per EPA AFS documentation)

0 SIP | 1 SIP under federal jurisdiction (FIP) | 3 Non-Federally Reportable | 4 CFC Tracking | 6 PSD | 7 NSR | 8 NESHAP (Part 61) | 9 NSPS | A Acid Precipitation | F FESOP (non-Title V) | I Native American | M MACT (Part 63) | T TIP (Tribal Implementation Plan) | V Title V.

Data note: the AIR_PROGRAM_CODE column also contains the values G and R, which are *not* in EPA's documented program-code list above; their meaning is not documented in the AFS download materials.

AFS_ACTIONS.csv

Compliance-monitoring and enforcement events, rolled up to plant level (1978–present).

Field	Description
AFS_ID / PLANT_ID	Plant identifiers
ANU1	Action number; uniquely identifies an action within a plant
NATIONAL_ACTION_TYPE	Two-character action code (inspection / enforcement). See note below
NATIONAL_ACTION_DESC	Text description of the action type
DATE_ACHIEVED	Date the action was completed (YYYYMMDD)
ALL_AIR_PROGRAM_CODES	All air programs associated with the action, space-delimited
PENALTY_AMOUNT	Civil penalty assessed or agreed, in dollars
RESULT_CODE	Result of stack tests / Title V reviews / actions
POLLUTANT_CODE	Pollutant associated with the action
ALL_VIOLATING_POLL_CODES	Pollutant(s) in violation, space-delimited
ALL_VIOLATION_TYPE_CODES	Violation type code(s)
KEY_ACTION_NUMBERS	Links to violation / FCE pathways (up to ten)
REGIONAL_DATA_ELEMENT_8	Region-specific data element
CREATION_DATE / DATE_RECORD_IS_UPDATED	Record metadata dates

NATIONAL_ACTION_TYPE translates region-specific codes to EPA national activity codes; the lead agency is indicated within each code's description. Per EPA, the most common codes are

FF, FS, FE, FZ, 1A, 5C (full inspections); EM, EO, ES, EX, PC, PO, PP, PR, PS, PX (partial inspections); and 1B, 2D, 6B, 7A, 7E, 7F, 8A, 8C, 9A (formal enforcement). The download contains **110 distinct** NATIONAL_ACTION_TYPE values and over 130 RESULT_CODE values; the full code lists are given in EPA's AFS documentation, and each row also carries its own NATIONAL_ACTION_DESC.

AFS_AIR_PRG_HIST_COMPLIANCE.csv

Quarterly compliance status per plant-program (FY2007-present). One row per plant-program-quarter.

Field	Description
AFS_ID	Plant identifier
AIR_PROGRAM_CODE	Regulatory program (same codes as AIR_PROGRAM)
HISTORICAL_COMPLIANCE_DATE	Quarter, YYQQ (Q1 = Jan-Mar, ..., Q4 = Oct-Dec)
HISTORICAL_COMPLIANCE_STATUS	Compliance status for the quarter (same codes as EPA_COMPLIANCE_STATUS)

AFS_HP_V_HISTORY.csv

Lifecycle of High Priority Violator designations. One row per HPV episode.

Field	Description
AFS_ID	Plant identifier
HPV_DAYZERO_TYPE	Lead-agency code at day zero (carried with HPV_DAYZERO_DESC)
HPV_DAYZERO_DESC	Text description of the day-zero type
HPV_DAYZERO_DATE	Date the plant entered HPV status
HPV_RESOLVED_TYPE	Resolution-action code (blank if unresolved; carried with HPV_RESOLVED_DESC)
HPV_RESOLVED_DESC	Text description of the resolution type
HPV_RESOLVED_DATE	Date the HPV was resolved

The *_TYPE code meanings are carried in the adjacent *_DESC columns in the data itself, and are listed in EPA's AFS documentation.